

As a basic introduction to these schools, a simple litmus test can be applied. Monte Carlo simulations are often used in financial planning contexts to gain a better understanding of the viability of a financial plan in the face of market and longevity risks. Suppose a Monte Carlo simulation identifies a retirement plan's chance of success as 90 percent. Both sides of the debate might accept this as the correct calculation from the software, but they will have dramatically different interpretations of what to do with this number.

For probability-based thinkers, a 90 percent chance is a more than reasonable starting point, and the retiree can proceed with the plan. It has a high likelihood of success, and that's enough for them. If future updates determine that the plan might be on course toward failure, a few changes, such as a small reduction in spending, should be sufficient to get the plan back on track.

Those identifying with the safety-first school, however, will not be comfortable with this level of risk, focusing instead on the 10 percent chance of failure. They make a distinction between essential expenses and discretionary expenses and seek a solution that practically eliminates the possibility of failure for meeting essential expenses. Jeopardizing success, they say, is only reasonable for discretionary expenses.

Financial service professionals and retirees should understand which school they most identify with and to what extent their own thinking might incorporate views from each school. Consumers of the financial services profession must understand whether they and their advisor are speaking the same language. Advisors able to communicate effectively from both sides will be more likely to deliver successful retirement income outcomes.

The Probability-Based School of Thought

Of the two approaches, the probability-based school of thought is probably the more familiar to the public and financial professionals. Its roots grow from research completed by California-based financial planner William Bengen in the 1990s. Bengen sought to determine the safe withdrawal rate from a financial portfolio over a long retirement. Though the term *safe withdrawal rate* uses the word *safe*, it is not part of the safety-first